FLOWOPERATIONS · OUTBOUND INFRASTRUCTURE

The Cold Email Infrastructure

Every moving part, what it does, and where it stands today

6,173	0	7	13
EMAILS SENT	CAMPAIGNS RUNNING	HUMAN REPLIES	AUTOMATIONS LIVE

Complete technical guide · live data pulled 2026-10-12
Internal — FlowOperations.ai

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01 · THE HONEST STATUS

Where things stand

This system finds outdoor-living contractors who are already spending money to get customers, builds each one a working branded tool on its own URL, and emails it to them free. The tool is the opener. A qualification call for the Marketing Trident is the goal.

> **Nothing has been sent in seven days**
>
> All four contractor campaigns are stopped — two COMPLETED, two PAUSED. The last email left on Oct 5. The cause is not a failure: the system ran out of leads. Smartlead holds 38 untouched leads against a daily capacity of 315. The Ramp Governor has been recording the reason every morning: "lead supply — only 38 untouched leads left".

The paradox worth understanding

All thirteen automations are healthy and ran successfully today. The study workflow is still reading practitioner blogs each morning and proposing new subject lines. Copy Forge has candidates staged and waiting. The reports still post four times a day. The ramp still calculates tomorrow's target. Every part of the machine works except the part that puts leads in the top.

Measure	Value	What it means
Emails sent, all time	6,173	Across four contractor campaigns
Campaigns running	0	Two completed, two paused
Days since last send	7	Last email Oct 5
Untouched leads loaded	38	Against 315/day capacity
Genuine human replies	7	0.11% of sends
— positive	2	Andrew (Aug 27), Ben (Sep 19)
— hostile or unsubscribe	4	Two said our claim about them was wrong
Calls booked	0	—
Real open rate	~14–18%	Not the 48.2% the dashboard shows
Automations live	13	All green, all ran today
Leads in master list	33,132	But only 917 fresh contractors

The three facts that matter

- The constraint is leads, not sending. We have the inboxes, the tools, the automation and the capacity. We do not have contractors to email.

- The copy never worked. Seven human replies from 6,173 emails is roughly 20x below the industry floor. Restarting with the same copy would reproduce the same result at greater volume.
- Two warm replies are sitting unanswered. Ben at Codex Construction asked "Ballpark?" on Sep 19. That is the most valuable thing this system has produced and it is three weeks cold.

02 · THE OFFER

What the system is for

The reverse lead magnet

A normal lead magnet asks a stranger for their email in exchange for a PDF. This inverts it: we do the work first, specifically for them, and hand it over before asking for anything. The contractor receives a finished branded asset with their company name on it, live on a real URL, whether or not they ever reply.

The reason is fulfilment risk. Earlier campaigns offered “reply and I’ll send you three things I found” — and that fulfilment failed essentially every time. A give-first offer cannot have a fulfilment gap, because the deliverable already exists when the email is sent.

Why an estimator for these contractors

The tool is not a creative choice. It is determined by where the contractor’s own customer gets stuck. For a homeowner considering a $15–60K patio, the stuck point is “I don’t know what this costs,” and they will not call three crews to find out. So the tool is a project estimator: pick square footage, material tier and features, get a typical investment range, and the contractor receives the lead with the full scope attached.

Where the buyer gets stuck	Tool form	Used here
”I don’t know what this costs”	Estimator / price calculator	Hardscaping, outdoor living
”I can’t reach anyone”	Booking page	Deck builders
”I don’t know if I qualify”	Eligibility checker	—
”How bad is my problem”	Diagnostic score	—
”I can’t picture it”	Visualiser	—

Signal over demographics

The biggest lever in cold email is not copywriting — it is reaching someone at the moment they show a real behavioural trigger. A contractor already paying for Google Ads, Meta ads or directory leads has admitted the problem with money, so there is no “do you have this problem” conversation to have. The system looks for evidence of spend before emailing anyone.

The priority stack

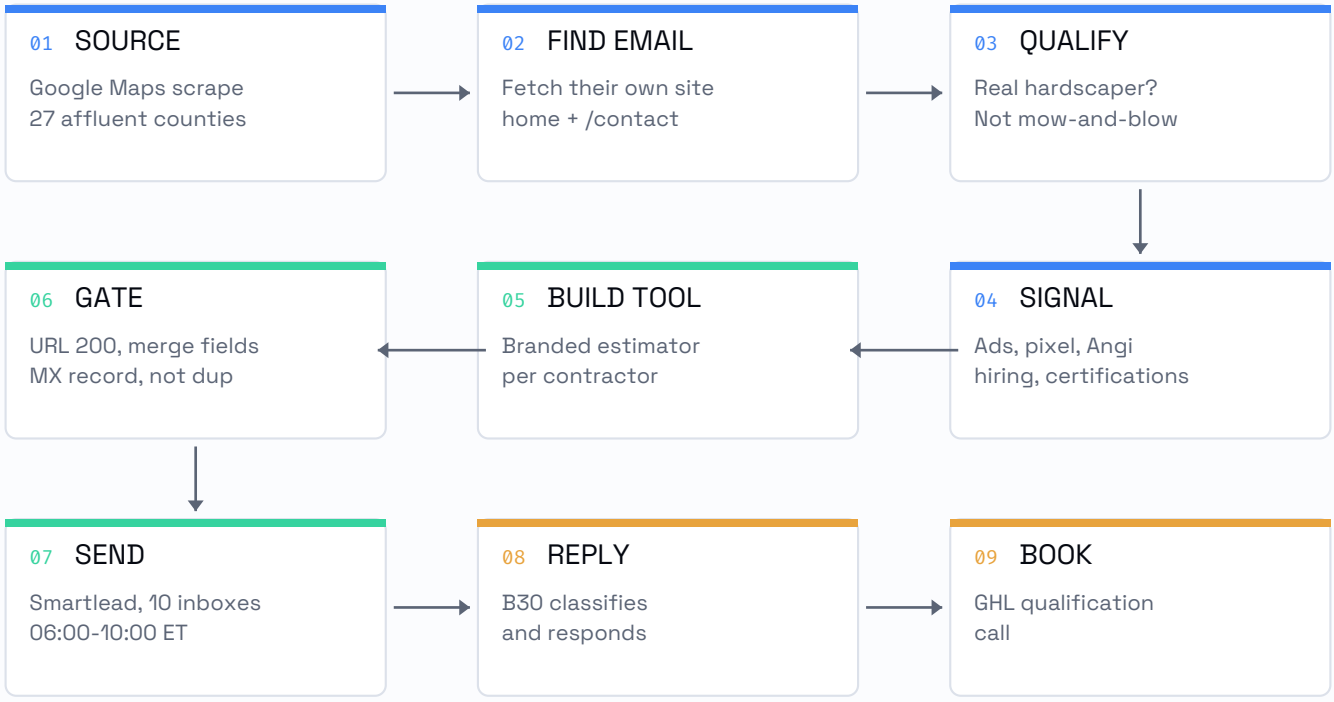
#	Lever	Meaning
1	Deliverability	Authenticate, warm, stay under complaint thresholds
2	Clean data	Verify every address before it enters a sequence
3	Relevance	Right person, right company, right moment
4	Offer	Specific value, not a vague ask
5	Personalisation	Amplifies everything above it; fixes nothing below it

> **What a season of this taught us**
>
> We fixed levels 1 and 2 thoroughly. Level 5 then turned out to be actively harmful: opening an email by telling a stranger what is broken about their business produced most of our hostile replies. Personalisation amplifies whatever it rests on, including a wrong premise.

03 · NINE STAGES

The machine, end to end

Every lead passes through nine stages. Blue stages decide who gets contacted, green stages prepare and send, amber stages handle what happens after. A lead failing any gate is dropped or paused — never sent in a broken state.



The stack

Layer	Tool	Role
Sourcing	Apify · compass/google-maps-extractor	Raw business records by category and county
Enrichment	Direct HTTP fetch (free)	Email discovery, qualification and signals from each site
Tools	Vercel · flowops-assets	~2,700 branded pages at /tools/{slug}
Sending	Smartlead	Campaigns, sequencing, scheduling, reply webhooks
Automation	n8n (exaltisgroup)	Thirteen workflows: replies, hygiene, ramp, study, reports
State	Apify key-value store + datasets	Ramp state, study memory, reply log, budget ledger
Intelligence	Claude (Sonnet)	Reply classification, copy generation, daily study

Layer	Tool	Role
CRM / booking	GoHighLevel	The qualification-call calendar
Reporting	Slack (Brutus) · Lovable	#cold-email reports; the war-room dashboard

04 · WHAT EACH STEP CHECKS

Every stage explained

Stage 1 — Sourcing

A Google Maps scrape by category and county, not metro. Hardscaping demand tracks lot size and home value, so the target list is affluent suburban counties with median home value above $500K — Bergen, Morris and Monmouth NJ; Westchester and Suffolk NY; Chester and Bucks PA; Fairfield CT; Fairfax and Loudoun VA; Collin TX; Douglas CO; Wake NC; Williamson TN. Search terms: hardscaping contractor, paver patio contractor, retaining wall contractor.

Measure	Last run
Places returned	1,460
Cost	$7.30 (hard-capped at $10)
Qualified leads produced	221
Cost per qualified lead	$0.033

Stage 2 — Email discovery

Free. For each domain the system fetches the homepage, then /contact, /contact-us and /about, and extracts addresses by pattern, preferring a named address over a role inbox. Hit rate on the last run was 70.9%. This stage caused most of our data-quality incidents.

> **What gets rejected here**
>
> Image and asset filenames (.avif, .webp, .png), template placeholders (example@mysite.com, you@company.com), third-party addresses scraped from font licences (team@latofonts.com, and the Google Fonts designer impallari@gmail.com), and malformed local parts.

Stage 3 — Layer-2 qualification

Google Maps category labels are unreliable in the trades — a lawn-mowing outfit is often listed as “landscaper.” So the system reads each site and requires all four of:

Gate	Passes when	Why
Real hardscaper	Hardscape vocabulary present: paver, retaining wall, seat wall, outdoor kitchen	43% of our first pool had none of it
Not mow-and-blow	Maintenance terms do not dominate (mowing, fertilisation, snow removal)	A $60K patio tool to a lawn service reads as spam
Portfolio present	A gallery or project page exists	Proves design-build, not maintenance
Maps quality	15+ Google reviews, 4.0+ rating	Below that is usually too small to buy

Stage 4 — Signal detection

The same HTML is searched for evidence the contractor already spends on acquisition. Scripts and styles are stripped first, and every signal requires real evidence rather than a keyword match — an early version matched icon class names and produced a 34% false-positive rate, which put false claims into eleven live emails.

Signal	Evidence required	Archetype
Meta ads	A real fbevents.js load or fbq('init') call	1 · paying to solve it
Google Ads	A real AW- conversion ID	1 · paying to solve it
Directory leads	An actual outbound link to angi.com, homeadvisor.com, thumbtack.com	1 · paying to solve it
Hiring	A careers/jobs link or "now hiring" text	2 · adding capacity
Certified	Belgard, Unilock, Techo-Bloc, EP Henry, ICPI badge	Revenue proxy (+2)

Stage 5 — Scoring and routing

Points	Criterion
+3	Currently spending money on the problem
+2	Signal detected under 14 days ago
+2	Manufacturer certification
+1	Named work email
-2	Role inbox only (info@, office@, sales@)

8+ routes to a hand-audited sniper tier, 5–7 to automated precision sends, under 5 is held. In practice every lead lands at 5–6 — there is no sniper tier yet, because it needs owner names and review mining, neither of which is built.

Stage 6 — Tool generation and deployment

Each qualified contractor gets a static HTML page generated from a template with a CONFIG block injected: company, city, domain, slug and webhook endpoint. Pages deploy to Vercel at `flowops-assets.vercel.app/tools/{slug}`. Estimators serve hardscapers; pages prefixed `book-` are booking pages for deck builders. The estimator shows a "typical investment range" and can read per-contractor pricing from `CONFIG.pricing`, so a contractor sets his own numbers rather than having a market rate published on his behalf.

Stage 7 — The send gates

- Tool URL returns HTTP 200 — added after an early email promised a link that 404'd.
- Every rendered merge field is populated — checked against every lead, never a sample. A 25-lead sample once came back clean while 1,275 leads were broken.
- Domain has an MX record — the address can actually receive mail.
- Not previously contacted in any campaign.

Stage 8 — Upload

Leads go to Smartlead with their custom fields: `signal, signal_noun, implication, gap_short, gap_detail, tool_url, city, score, tier, certs`. These per-lead fields are the only part of the copy that can be safely changed on a live campaign.

Stage 9 — Sending

06:00–10:00 ET, Monday to Friday, three to five minutes apart, across ten inboxes at a 35/day per-inbox cap. The window came from our own data, not from a playbook: 06:00 sends opened at 59.0% against 38.2% at 08:00. Stop-on-reply is on. Follow-ups use a blank subject so Smartlead threads them natively.

Stage 10 — Reply handling (B30)

Every reply fires a webhook into n8n workflow B30, which runs in this order:

#	Node	Does
1	Webhook + 200 OK	Acknowledges instantly so Smartlead never retries
2	Parse and Lookup	Identifies the prospect from sl_lead_email / to_email
3	Claude Classify	POSITIVE_HIGH / LOW, NOT_NOW, UNSUBSCRIBE, OOO, BOUNCE, SPAM
4	Insert Reply Row	Logs to the fo-replies dataset
5	Handle Unsubscribe	Adds to the global block list — suppressed across all campaigns
6	Auto-Send Guardrail	Safety checks before anything is sent
7	Call Push	On a positive: sends tool link + booking link, once, idempotently
8	Research + Slack card	Posts the reply, classification and context to #cold-email

A reply arriving in under six minutes is treated as a machine. That one rule is what separates our 7 genuine replies from the 31 the dashboard counts.

Stage 11 — Booking

A GoHighLevel calendar link. When a call is booked, a workflow threads the booking into the original Slack reply card. This is the weakest stage: no qualifying questions, no reminders, no no-show recovery — and it has never been exercised, because no call has been booked.

Stage 12 — Tool lead capture

When anyone submits a deployed tool, the page posts to `/fo-tool-lead`, which logs the submission and posts a Slack card to FlowOps.

> **The lead still never reaches the contractor**
>
> The workflow has four nodes — webhook, acknowledge, log, Slack — and sends nothing to the contractor. The email tells him “the request lands with you instantly — your lead, not a shared list.” If a homeowner used any of the ~2,700 deployed tools today, the contractor would never know. This was identified on Sep 21 and is still open.

05 · THIRTEEN WORKFLOWS

The automation layer

The system runs itself on a fixed daily clock. Everything that decides volume runs before the send window opens at 06:00. Every workflow below ran successfully today.

ET	Workflow	What it does
03:30	Daily Study	Reads practitioner blogs, critiques our live config, proposes changes
04:00	Copy Forge	Generates and lints new subject candidates against the 2026 rules
05:00	Split-Test Referee	Scores variants and recommends (auto-swap deliberately disabled)
05:00	Inbox Onboarding	Ramps new inboxes on an age curve, strips signature images
05:15	Lead Hygiene	Junk and MX checks, plus the bounce circuit breaker
05:30	Phase Optimizer	Rescores the ten pipeline stages and flags the weakest
05:45	Ramp Governor	Sets volume = min(curve, inbox capacity, lead supply)
06:00	SEND WINDOW	Smartlead fires until 10:00 ET — currently nothing to send
09:00	Morning Briefing	The elaborate report: what is going out, what went, what worked
14:00	Midday Pulse	Window report, 09:00 → 14:00
18:00	Evening Report	Window report, 14:00 → 18:00
23:59	Night Recap	Day close plus tomorrow's ramp target
hourly	Budget Guard	Caps spend; every paid call checks in here first
on reply	B30 Reply Pipeline	Classifies and responds to every inbound reply
on submit	Tool Lead Capture	Fires when anyone submits a deployed tool
on demand	Brutus Ad-hoc	Pushes a message to Slack when asked

Design rules the automation follows

- Governors, not counters. The ramp sets volume to the minimum of the growth curve, warmed inbox capacity and remaining lead supply, and freezes entirely if bounce passes 2%.
- Model calls live in HTTP nodes. n8n aborts Code nodes at 60 seconds; three workflows failed silently for four days before this was found.
- Safety rails on deletion. Hygiene aborts if any filter flags more than 3% of a campaign — written after a cleanup deleted 177 legitimate leads.
- Judge on recent data. The bounce breaker reads the last 200 sends after the most recent cleanup, so a historical spike cannot lock a campaign off forever.
- Recommend, do not rewrite. The referee no longer auto-swaps variants, because every sequence write re-queues already-contacted leads.
- Cost is logged. Token usage for every Claude call is written to `llm-usage-YYYY-MM-DD`, because n8n discards Anthropic's usage block by default.

06 · INBOXES, DOMAINS, CAPACITY

Sending infrastructure

Ten mailboxes across two domains. All are mature — 169 to 176 days old — and all report 100% warmup reputation with warmup still running.

Domain	Inboxes	Personas
flowoperations.org	5	lilly, priya, ryan, ashley, nealt
flowopsleadgen.com	5	roddy, mason, john, carly, neal

Eight of the ten send as fictional people with invented titles; two use Neal's own name. Andrew Hardison replied to "John Carter," who does not exist, and would have met Neal on the call. Common practice, but an open decision — recorded here rather than quietly changed.

Authentication

Record	flowoperations.org	flowopsleadgen.com
SPF	include:_spf.google.com ~all	include:_spf.google.com ~all
DKIM	Google selector, present	Google selector, present
DMARC	p=none	p=none
MX	Google Workspace	Google Workspace

Authentication is valid. DMARC at `p=none` is the weakest policy and is worth raising to quarantine, but on its own it does not cause spam placement.

Capacity arithmetic

	Value
Inboxes / domains	10 across 2
Per-inbox cap	35 / day (cut from 50, which exceeded the 20–30 consensus)
Effective capacity	350 / day
Safe send (90% margin)	315 / day
Leads available today	38
Needed for the 5,000/day goal	~167 warmed inboxes across ~60 domains

> **Capacity is not the bottleneck and has not been for weeks**
>
> The ramp is on day 48 of a 128-day curve to 5,000/day. It could send 315 emails today. It has 38. Adding domains right now would buy nothing — the ceiling that binds is lead supply, and after that, copy that earns replies.

New-inbox ramp curve

Inbox age	0–2 d	3–6 d	7–13 d	14–20 d	21+ d
Daily cap	5	10	20	28	35

Graduation to live campaigns needs 14+ days, warmup reputation at or above 90%, and active warmup. Existing inboxes were seeded with their real creation dates so the ramp did not mistake five-month-old mailboxes for new ones.

07 · AND WHAT THEY REALLY MEAN

The numbers

By campaign

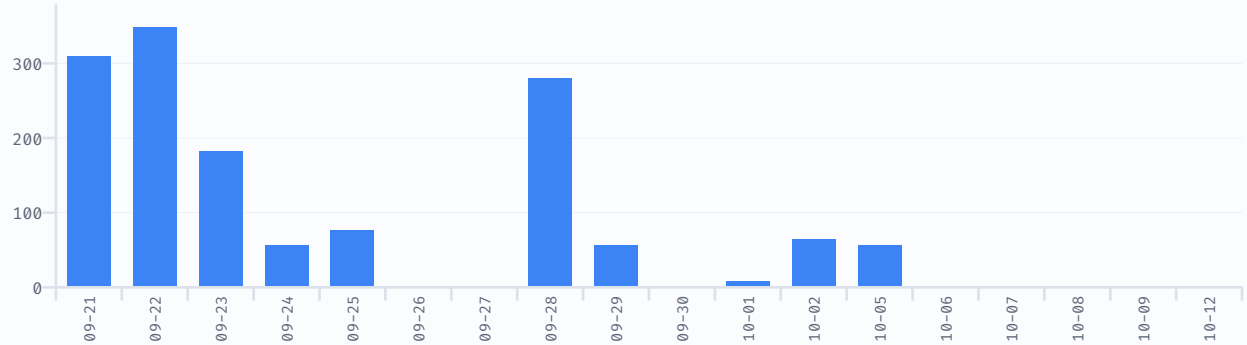
Campaign	Status	Sent	Replies	Human	Bounce	Leads left
Hardscaping	COMPLETED	3,370	12	2	0.8%	0
Deck Builders	PAUSED	1,541	12	3	3.8%	356
Hardscape v2	PAUSED	225	1	1	7.1%	37
Outdoor Living	COMPLETED	1,037	2	1	0.4%	0

“Replies” is what Smartlead counts. “Human” is what survives the six-minute filter. The gap between those two columns is the single most important thing on this page.

Daily volume

Emails sent per day, Sep 21 — Oct 12

← seven days of total silence

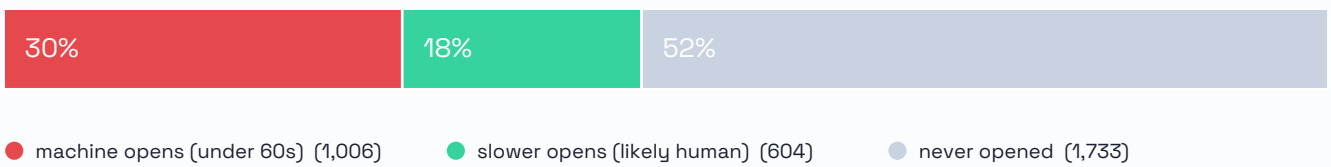


The gaps are weekends; the run of zeros from Oct 6 is the list running dry.

The open rate is not measuring humans

Only Hardscaping has open tracking on, so it doubles as the reputation monitor. Smartlead reports 48.2%. The pixel data says otherwise.

What Smartlead calls a 48.2% open rate, broken down (3,343 delivered)



Evidence	Finding
Time to first open	Median 24 seconds. 62% of all opens land inside the first minute — that is a security gateway fetching images on arrival, not a person.

Evidence	Finding
Clicks	0 clicks on 1,610 opens, with click tracking enabled. The entire point of the email is a link. Machines fetch pixels; they do not follow links.
Timing	Opens cluster inside the 06:00–10:00 send window rather than spreading across the day.
Known humans	Of the people who replied — certain readers — 27% registered no open at all. Their client blocked images.

Figure	Value	Status
Smartlead dashboard	48.2%	Fiction
Excluding opens under 60s	18.1%	Plausible floor
Excluding opens under 5 min	14.2%	Conservative floor

Treat 15–20% as the real number. It is an ordinary cold-email result — enough inbox placement that copy, not deliverability, is the binding constraint.

Every genuine human reply, ever

Date	Campaign	Who	Read	What they said
08-24	Hardscaping	fairviewlandscapinginc.com	—	first reply, pre-classifier
08-27	Hardscape v2	hardisonlandscaping.com	POSITIVE	"Sure, I'd be interested in seeing it."
09-09	Hardscaping	friesenlandscaping.com	Hostile	"Get lost!"
09-09	Deck Builders	dynamicdecksolutions.com	Hostile	"You must be blind, you can book an appointment with me thru my portal."
09-10	Deck Builders	gmail.com	Hostile	"...you're not even correct about your introduction"
09-14	Outdoor Living	graveltogold.com	Unsub	"STOP"
09-19	Deck Builders	codexconstruction.com	POSITIVE	"Ballpark?"

What the replies say

Both positives came from neutral emails — Andrew from the signal variant, Ben simply asking the price. Two of the hostile replies told us our claim about their business was factually wrong. An automated copy analysis independently measured the neutral signal variant beating the diagnostic ones. Small numbers, but every piece of evidence points the same way: stop criticising people.

Against the industry

Metric	Average	Good	Exceptional	Us
Reply rate	2–3%	4–6%	8–10%	0.11%
Positive reply	1–2%	2–4%	4–5%	0.03%
Meetings per 1,000	5–8	8–15	15–30	0
Bounce rate	3–5%	under 3%	under 1%	1.7% — 2.6%

Bounce is the one metric at exceptional level — the list-hygiene work paid off. Everything else is roughly 20x below the floor.

> **The arithmetic that should drive every decision**
>
> The goal is 3–5 calls a day. At top-quartile conversion (1.5% booked), 300 emails a day — what the current ten inboxes already support — produces about 4.5 meetings a day. We do not need 5,000 emails a day to hit the target. We need the copy to work at a rate thousands of operators already achieve. The whole game is the gap between 0.11% and 2%.

08 · WHY WE RAN OUT

The lead engine

A full audit of every lead ever gathered produced a master list of 33,132 unique contacts, merged from all 24 Smartlead campaigns, 37 local CSV exports and the Apify datasets, deduplicated by email.

Source	Records	Unique
Smartlead, all 24 campaigns	24,039	23,684
Local CSV exports (37 files)	44,330	31,339
Apify fo-leads-scored	130	0 new
Master list after dedupe	—	33,132

But the usable figure is far smaller

Segment	Count	Note
Total unique leads	33,132	Everything ever gathered
Already contacted	5,183	Cannot be re-used
Fresh contractor leads	917	The only niche with tools and copy built
Loaded in Smartlead, untouched	38	What the sender can actually reach today
Other niches (finance, insurance, etc.)	~27,000	No tools, no copy, many over a year old

> **This is the whole explanation for the silence**
>
> 917 fresh contractor leads is about three days of sending at 300/day. We spent them, and the campaigns completed. The 33,132-row master list looks like abundance and is not: most of it is other niches with no tool built, no copy written, and addresses old enough that sending to them would reproduce the 11.9% bounce incident.

What refilling actually costs

The last scrape returned 1,460 places for $7.30 and produced 221 qualified leads — 3.3 cents each. At 250 sends a day, supply costs roughly $8.60 a day. That is the entire marginal cost of keeping the machine fed. Every other component is already paid for.

09 · INCIDENTS AND RULES

What we learned

Every rule in this system was paid for. Each row records the incident, what it cost, and the rule it produced.

Incident	What happened	Rule it produced
Signal false positives	Substring matching flagged icon classes and CSS as "buys directory leads." 34% false; 11 emails made a false claim.	A signal that becomes a factual claim needs real evidence, never a keyword.
Duplicate sends	Rewriting a live sequence re-queued contacted leads. 86 contractors got two cold emails hours apart.	Never write to a live campaign's sequence. Ship copy changes as a new campaign.
Dead auto-responder	Smartlead's webhook puts OUR mailbox in from_email and the prospect in to_email. Every reply was attributed to ourselves.	Read sl_lead_email / to_email. Verify the payload, not the field name.
Lawn companies emailed	43% of the pool had no hardscape vocabulary. 102 sends went to maintenance outfits.	Verify the trade from the site. Category labels lie.
11.9% bounce	The email scraper harvested image filenames and template placeholders.	Filter .avif/.webp and placeholder domains at the point of extraction.
177 good leads deleted	A cleanup filter matched real first names (Joe, John) and flagged contractors with a second domain.	Deletion filters are allow-list narrow. Real people are named Joe.
1,275 broken emails nearly sent	A restore uploaded leads without the merge fields the sequence renders.	Scan every lead after an upload. A 25-lead sample hid it.
Silent workflow failures	Three workflows hit n8n's 60-second Code-node limit for four days.	Model calls go in HTTP nodes; cap pagination.
Reputation decay	Open rate fell from 56.8% to 26.7% on two young domains at 50/inbox.	The volume ceiling is set by domains. Cap at 35/inbox.
Hostile replies	Openers criticised the recipient's business, sometimes wrongly.	State only what was observed, or make no claim at all.
Incoherent copy	Subject, body and merge fields were edited separately and stopped agreeing.	Copy lives in three places. Check them against each other before sending.

10 · OPEN DEFECTS

What is broken now

#	Defect	Scope	Severity
1	Campaigns stopped — no leads loaded; nothing has sent in 7 days	Whole program	Critical
2	Tool submissions never reach the contractor; the email promises they do	All ~2,700 tools	Critical
3	Ben at Codex Construction asked "Ballpark?" on Sep 19 — still unanswered	1 warm lead	Urgent
4	Subject lines break the 2026 rules (fake "Re:", questions, ellipsis, "free")	All step-1 copy	High
5	"Rather than send you a list of problems" refers to nothing after the opener rewrite	Hardscaping body	High
6	Body promises "plus a monthly payment" the tool no longer shows	Hardscaping body	High
7	Deck Builders at 3.8% bounce, Hardscape v2 at 7.1% — both above the breaker	2 campaigns	Medium
8	Tool type does not always match the copy (booking page sent with estimator copy)	119 leads	Medium
9	Auto-responder sends a link on a question instead of answering it	B30	Medium
10	Booking flow has no qualifying questions, reminders or no-show handling	GHL	Medium
11	LLM calls go direct to Anthropic; house rule is OpenRouter	3 workflows	Low
12	DMARC still p=none on both domains	2 domains	Low

Why defects 4–6 are still live

Smartlead recreates every sequence variant on write — even when each variant's existing ID is passed back. Tested directly: IDs 7453213… became 7711429… on a write that changed nothing. Any edit to live copy therefore re-queues already-contacted leads and sends them a second cold email. The only safe fix is a new campaign. Per-lead merge fields are the exception, which is how the openers were fixed in place on Sep 14.

The silver lining in defect 1

With every campaign stopped and no leads queued, the duplicate-send risk that blocked the copy rebuild for a month is gone. A clean campaign can be built and loaded with zero chance of double-emailing anyone. The outage is the window.

11 · IN ORDER

How to restart it

When	Step	What
Now	Answer Ben	Reply to "Ballpark?" with a straight number. Second positive the system ever produced; three weeks cold.
Now	Make the promise true	On every tool submission, email the lead to the contractor. The payload already carries the domain. Fix the Slack card copy to match.
Step 1	Rebuild the copy	One clean campaign: compliant subjects, a body with no orphaned lines and no monthly-payment promise, plus a pre-send coherence check that subject, body, merge fields and tool type agree.
Step 2	Refill the list	~$8.60/day of Google Maps scraping keeps 250 sends/day fed. Route by tool type so booking-page leads get booking copy.
Step 3	Restart small	150–200/day into the new campaign only. Hold there until reply rate clears 2% on a few hundred sends.
Step 4	Daily guard	Render one real email per campaign each morning and fail loudly on unresolved tags, rule violations, tool mismatch, or any claim the tool cannot back.
Step 5	Fix the reply path	Answer questions before sending links. Add qualifying questions, reminders and no-show recovery in GHL.
Step 6	Only then, scale	As reply rate holds above 2%, let the ramp raise volume. Spread across new domains, not more mailboxes on the existing two.

Open decisions

- Peer proof. A nameable client and a result worth standing behind unlocks the strongest template for a tight trade. Endurance Hardscaping is the obvious candidate.
- Personas. Keep the eight fictional senders, or move to real names.
- DMARC. Raise both domains from p=none to p=quarantine (needs registrar access).
- Recurring scrape spend. ~$8.60/day to keep supply ahead of a 250/day send rate.
- The other 27,000 leads. Re-verify and build tools for a second niche, or leave them.

> **An honest limit**
>
> Seven human replies from 6,173 sends is a small sample. This plan removes defects that can be proven to exist and restarts at a volume where learning is cheap. Whether it produces a materially higher reply rate is a hypothesis to test, not a promise — but the test is now cheap, because the system measures itself every day and the infrastructure is already paid for.

A · IDS AND ENDPOINTS

Reference

Campaigns

ID	Name	Role
3842105	Hardscaping	Tracked; reputation monitor; now completed
3845617	Deck Builders	Booking-page offer; paused on bounce
3863467	Hardscape v2	Compliant copy; paused, junk list
3848711	Outdoor Living	Four-template test; completed

Workflows (n8n · exaltisgroup.app.n8n.cloud)

n8n ID	Workflow	Runs
TH6uS1Phh8jZCmTG	B30 Reply Pipeline	on every reply
oXeFIbzjw458LXdN	Tool Lead Capture	on every tool submission
9yaBaX37Szw099yV	Daily Study	03:30
Tjnp1M4DRhng6TZM	Copy Forge	04:00
hNoZZPVG7burR5W4	Split-Test Referee	05:00
eMFl89iztFbzN32p	Inbox Onboarding	05:00
AuaSqfQIFjuE87RJ	Lead Hygiene	05:15
UeeKFQ7ziTzl9mzT	Phase Optimizer	05:30
bDQW15jNVtSQs2FW	Ramp Governor	05:45
D5nP8X4CUp2QU6RU	Reports 4x daily	09:00 14:00 18:00 23:59
ZeLjNDS2L5U8YiUb	Budget Guard	hourly
uYDhBPX0MQMSEo6z	Daily Digest	daily
rBT2hx6BW9rE7uhr	Brutus Ad-hoc	on demand

Webhooks

```
https://exaltisgroup.app.n8n.cloud/webhook/
  brutus-reply       every Smartlead reply
  fo-tool-lead       every tool submission
  fo-report-now      force a report  {"kind":"morning|midday|evening|night"}
  fo-study           run the daily study
  fo-hygiene         run lead hygiene
  fo-ramp            run the ramp governor
  fo-inboxes         run inbox onboarding
  fo-budget-guard    {"action":"status"|"check_and_commit", ...}
```

Smartlead API behaviour worth knowing

Behaviour	Consequence
Sequence writes recreate every variant, even with IDs passed back	Live copy cannot be edited safely
Reply webhook: from_email = our mailbox, to_email = prospect	Read sl_lead_email or to_email
Tracking flags normalised on write (DONT_TRACK_EMAIL_OPEN → DONT_EMAIL_OPEN)	Checks must match both spellings
track_settings: [] means track everything	An empty array is not "no tracking"
statistics.email_subject is the rendered subject	Attribute variants by seq_variant_id
Leads endpoint maximum limit is 100	A limit of 500 returns HTTP 400
The /leads/ account endpoint requires an email parameter	Enumerate per campaign instead
Follow-up bodies must sit inside seq_variants	A step-level body saves an empty step
New campaigns do not inherit webhooks	Create one per campaign, id: null, all 7 categories
DELETE on a lead returns an empty body	Verify by re-counting, not by response
Two bulk scripts at once return 429	Serialise; ~0.2s between per-lead calls
max_new_leads_per_day = 0 is rejected	Use 1 to drain a campaign
The campaign /schedule endpoint returns an empty body	Read the schedule off the campaign object

State (Apify key-value store gpumCUeww5uaQXYr5)

`ramp-state` · `inbox-state` · `inbox-registry` · `split-test-state` · `phase-optimizer-state` · `study-latest` · `study-memory` · `copy-forge-memory` · `copy-forge-candidates` · `lead-hygiene-state` · `hygiene-cleaned-at` · `llm-usage-YYYY-MM-DD` · `rsnap-{morning|midday|evening|night}`

Datasets: `fo-replies 2X1f0gicbZsO3fOT7` · `fo-voc 36I6b2kiUpu19bd3r` · `fo-leads-scored XmbdosdmwpZD2AenV`

Elsewhere

Thing	Where
Tool pages	`flowops-assets.vercel.app/tools/{slug}`
Dashboard	`trident-war-room.lovable.app/cold-email-august-2026`
Slack reports	`#cold-email (C0BRH7PBC14)`
Booking link	`api.leadconnectorhq.com/widget/bookings/flowqualificationcall`
Master lead list	`~/Desktop/FlowOps_Master_Lead_List.csv`
Scraper	`Apify compass~google-maps-extractor`

B · WHAT MOVED AND WHERE

Cloud migration

On 20–23 October the system was moved off this Mac. Nothing about how the cold email engine works changed — what changed is that losing the laptop no longer loses the work.

What was already cloud

Most of the engine never lived on the Mac: Smartlead does the sending, n8n runs all thirteen automations on a schedule, Apify holds the scraper and state, Vercel serves the 2,558 tool pages, and Slack and GoHighLevel handle reporting and booking. Those ran untouched throughout.

What moved

What	From	To	Status
Tool page source · 2,641 files	~/Desktop/flowops-assets	GitHub flowops-assets (private)	Done
Ops scripts · 122 files	~/Desktop/council	GitHub flowops-ops (private)	Done
Master lead list · 33,132 rows	Desktop CSV	Apify dataset fo-master-leads	Done
29 API keys in source	29 .py files	~/.config/n8n-creds/flowops-core.env	Done
Auto-deploy on push	manual CLI	Vercel ← GitHub	Blocked
Generator scripts running unattended	this Mac	n8n / scheduled runner	Deferred

The security finding that justified doing this first

Twenty-nine live API keys — Smartlead, Anthropic, OpenRouter and Apify — were sitting in plaintext across 29 Python files, including a core/ subdirectory and a .bak file. Four distinct secrets in total. All are now read from the environment, and a secret gate ran over every tracked file before either repository was pushed. Both repos are private and verified clean.

Because those keys sat on disk for months, cleaning the source stops future exposure but does not undo past exposure. Rotating all four is cheap insurance — Anthropic and OpenRouter first, since they bill directly. New values go in the same env file and every script picks them up with no code change.

Where everything lives now

Resource	Location
Tool page source	`github.com/NealFlowOperations1993/flowops-assets`
Ops scripts	`github.com/NealFlowOperations1993/flowops-ops`
Live tool pages	`flowops-assets.vercel.app/tools/{slug}`

Resource	Location
Master lead list	Apify dataset cMhQiJNz6imqqTNOb (fo-master-leads)
Automations	exaltisgroup.app.n8n.cloud — 13 workflows
Campaigns	Smartlead · 4 contractor campaigns
Reporting	Slack #cold-email (C0BRH7PBC14)
Credentials	~/.config/n8n-creds/flowops-core.env (local, chmod 600)

Two things still tied to the Mac

1. Vercel auto-deploy. The Git connection fails because the Vercel account (neal-2540 / neal@flowoperations.ai) has no GitHub Login Connection — confirmed directly against the Vercel API, which reports it as NONE. This is an account-level OAuth link, separate from the GitHub integration. Until it is added at vercel.com/account/login-connections, deploys still run from the laptop with `npx vercel@latest deploy --prod --yes`. Note the project is owned by the team `flow-operations`, not the personal account.

2. The generator scripts. The 28 scripts in flowops-ops still only run when invoked on this machine. Deferred deliberately: automating the production of emails that currently earn a 0.11% reply rate is not worth one to two days of work until the copy rebuild proves out. The thirteen n8n workflows already cover reply handling, hygiene, ramp, study and reporting — what is missing is only the lead-generation half.

C · HOW TO PICK THIS UP

The cold email project

Cold email now has its own Claude Code project, separate from the general ~/Desktop workspace and its 1,623 mixed sessions.

Opening it

```
cd ~/Desktop/council
claude
```

Claude Code scopes a project to its working directory, so any session started from here is a cold-email session with its own memory and history. The directory keeps the name `council` because nine scripts hardcode that path; on GitHub it is `flowops-ops`.

What the project contains

Path	What
`CLAUDE.md`	Project instructions — loaded automatically every session
`docs/`	This guide, plus the restart plan
`core/`	Shared modules: smartlead_io, llm, signal_enricher, website_redo
`*.py`	28 ops scripts — scrape, qualify, score, generate, verify, upload, report
`.gitignore`	Excludes *.env, *.csv, __pycache__, *.bak

Before running anything

```
set -a; source ~/.config/n8n-creds/flowops-core.env; set +a
```

Every script reads its credentials from the environment now. Without this line they will raise a KeyError rather than silently using a stale key — which is the intended behaviour.

What has not changed

> The migration protected the work. It did not fix the work.
>
> All four campaigns are still stopped and nothing has sent since 5 October. The engine is starved of leads, not broken. Seven human replies from 6,173 emails is roughly twenty times below the industry floor, and that is a copy problem. The restart sequence in Section 11 is unchanged: answer Ben, make the tool promise true, rebuild the copy as one clean campaign, refill the list, restart small at 150–200/day, and only scale once reply rate clears 2%.

One advantage the outage created

Smartlead recreates every sequence variant on write, which is why editing live copy would have re-queued already-contacted leads and sent them a second cold email. With every campaign stopped and nothing queued, that risk is gone. A clean campaign can be built and loaded now with no chance of

double-emailing anyone.

First three moves

	Move	Why
1	Reply to Ben at Codex Construction	He asked "Ballpark?" on 19 September. Second positive reply the system ever produced.
2	Wire the contractor notification	Tool submissions reach FlowOps but never the contractor, while the email promises they do. One workflow, four nodes, fixes all 2,558 pages.
3	Rebuild the copy as a new campaign	Compliant subjects, no orphaned lines, no promise the tool cannot keep, plus a pre-send coherence check.